

EE 10510: FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING

PRACTICE SHEET-1

- 1. Explain the V–I characteristics of a PN junction diode.**
 - Describe the forward and reverse bias regions and the concept of breakdown voltage.
 - 2. Differentiate between NPN and PNP transistors.**
 - Mention current directions, symbol differences, and typical applications.
 - 3. Define and explain the terms α (alpha) and β (beta) in a BJT.**
 - Derive the relationship between α and β .
 - 4. Explain input and output characteristics of an NPN transistor in CE Configuration**
 - 5. Describe the working of a diode as a half-wave and full-wave rectifier.**
 - Include the circuit diagram and explain the output waveform qualitatively.
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◆ Numerical Questions (5)

6. Diode Current Calculation:

A silicon diode has a forward voltage drop of 0.7 V when conducting. Find the diode current if the circuit has a 10 V supply and a 2 k Ω series resistor.

Formula:

$$I = \frac{V_{supply} - V_{D_}}{R}$$

7. Reverse Saturation Current:

A diode has a saturation current $I_s = 10^{-12}$. Find the diode current when forward biased at 0.7 V and temperature 300 K.

Use: $I = I_s(e^{V/(\eta v_t)} - 1)$, where $v_t = 26\text{mV}$ and $\eta = 1$.

8. Transistor Current Relations:

A transistor has $\alpha = 0.98$ and a collector current $I_C = 4.9\text{ mA}$. Find the emitter current (I_E) and base current (I_B).

Use: $I_C = \alpha I_E$ and $I_E = I_B + I_C$.

9. Given that $\beta=120$ and $I_C=2\text{mA}$. Find I_E, I_B and α .

10. In a common-emitter transistor circuit:

- $V_{CC}=12\text{V}$
- $R_C=2\text{k}\Omega$
- $\beta=100$
- $I_B=30\mu\text{A}$

Find:

1. The **collector current** (I_C)
2. The **collector-emitter voltage** (V_{CE})